Research Statement

Mahta Ghafarianghadim

I am an applied microeconomist who studies how environmental exposures, health behaviors, and public policies shape health and health equity. Because researchers cannot randomly assign pollution, health behaviors, or policy exposure, causal relationships in this area are hard to identify. My work addresses this by using instrumental variables and difference in differences designs with restricted access mortality and administrative data to study air pollution, smoking, early childhood outcomes, and reproductive health policy. Two commitments tie this work together: a focus on the populations who bear the highest costs of these exposures and policies, and care in getting the identification right so that the findings are reliable enough to inform regulation and law. This focus on policy relevance was recognized with the Dan E. Sweat Dissertation Fellowship.

My job market paper, "Air Pollution and Cardiovascular Mortality: Evidence from Thermal Inversions in the U.S.," estimates the causal effect of fine particulate matter (PM2.5) on cardiovascular mortality, the leading cause of death in the United States. Pollution is not randomly assigned: places with more pollution differ in economic activity and health care access, and pollution is hard to measure well. To address this, I use thermal inversions, a weather event in which a layer of warm air sits above cooler air and traps pollution near the ground. Since inversions are driven by weather rather than local economic or health conditions, they raise pollution in a way that is not tied to other causes of death. Thermal inversions are increasingly used in this way, but studies build the measure differently and rarely check how these choices affect the estimates. I show that the common way of measuring inversions misses about 30 percent of U.S. counties, mostly at higher elevations, while measuring temperature relative to the local ground covers every county and gives a much stronger first stage. Using this measure, I find that a 1 $\mu g/m^3$ increase in PM2.5 raises cardiovascular mortality by 3.8 percent in the average county. Weighted by population, this implies an annual cost of about $124 billion using the EPA's value of a statistical life. The results pass placebo tests, and I find no evidence of mortality harvesting. The effects are largest for Black individuals, women, the elderly, and low income and rural counties. These results suggest that current air quality standards under the Clean Air Act may understate the health costs of PM2.5 and that regulation should account for who bears these costs.

A second project, "Air Pollution and Early Childhood Outcomes: Evidence from Georgia Childcare Providers" (with Garth Heutel), studies whether pollution keeps preschool age children out of childcare, a channel that researchers have largely overlooked. Using restricted administrative data from the Georgia Department of Early Care and Learning, obtained through a partnership with the Georgia Policy Labs, and thermal inversions as the instrument, we find that a one unit increase in weekly county average particulate pollution decreases childcare attendance by 2.5 percentage points, with larger effects for Black and Hispanic children and children with working caregivers. Environmental conditions affect not only health but also families' access to

important social institutions. The paper is an NBER working paper and is under review at the Journal of Environmental Economics and Management (JEEM).

A third project, "Mental Health and Functional Consequences of Smoking: A Causal Approach Using Misreporting Corrections," examines how smoking affects mental health and daily functioning. Smokers differ from nonsmokers in ways that affect health, and people often underreport smoking in surveys, so even instrumented estimates are biased. I address both problems by combining a lagged cigarette tax instrument with a bivariate probit model that separately estimates the probability of being a true smoker and of reporting honestly. Preliminary estimates show that smoking raises the probability of poor mental health and functional limitations by more than standard estimates suggest, which means the mental health benefits of tobacco control policies are larger than previously thought.

Outside my dissertation, I am developing a paper titled "The Effect of Targeted Regulation of Abortion Providers (TRAP) Laws on Food Insecurity in the U.S." TRAP laws restrict abortion access through legal requirements placed on providers, a clear case of state laws shaping wellbeing beyond their stated purpose. Using a staggered difference in differences approach over 2010 to 2019, I estimate whether these restrictions impose financial costs on households beyond direct healthcare access, a question that has become more pressing since the Dobbs decision returned abortion regulation to the states.

My research agenda extends this work in two directions. The first moves to environmental health domains where the medical literature shows associations, but credible causal estimates are missing. One project examines how nitrate contamination in drinking water affects thyroid health. Nitrate is the most common chemical contaminant in the U.S. drinking water supply and is consistently linked to thyroid disease, but no economics paper has produced a causal estimate, and such estimates would help policymakers weigh the full cost of contamination when they set and enforce drinking water standards. The second direction follows from my job market paper: many natural experiment designs rest on construction choices that researchers rarely test, and I bring the same diagnostic approach to other instruments and policy settings. I will carry out some of these projects on my own, as I did with my job market paper, and others with coauthors and agency partners, as in my childcare work, since some questions need one person's focus and others need shared data and expertise.

This agenda also lends itself to working with students. Much of it relies on public data, such as mortality records, air and water quality monitoring, and state policy databases, and I plan to involve undergraduate and PhD students as research assistants and advise theses that ask how environmental conditions or public policies affect health. I plan to support this work with external funding early in my appointment. Through the childcare project, I gained experience developing proposals to the American Lung Association and the Caplan Foundation. My next step for the nitrate project is to seek external grant funding that brings students into the research and to build partnerships with state health and environmental agencies that hold the data these questions require.